HUJI Research Monitor

Monthly Digest · May 2026 · May 01, 2026

EXECUTIVE SUMMARY

Hebrew University researchers are advancing groundbreaking solutions across critical sectors. Innovations include a novel AI for personalized Parkinson's management, a sustainable bio-pesticide revolutionizing agriculture, and promising drug discovery for autism spectrum disorder. Additionally, new diagnostic tools are emerging to optimize existing therapies and address neurodevelopmental delays. AI-powered health tech targets global health disparities, demonstrating broad commercial potential.

#1

31/50

A feasibility open-label clinical trial utilizing second-generation artificial intelligence based on the constrained-disorder principle in patients with Parkinson's disease.

Main Researcher: Ilan Y

[Software/AI] [Neuroscience] [Clinical] [Medical Device]

"Next-gen AI for Parkinson's disease offers personalized management and therapeutic potential."

WHY NOW

There is an urgent need for precision medicine in Parkinson's disease, and AI integration with existing or new wearables creates a scalable solution for personalized care.

COMMERCIAL ANGLE

Commercialization could focus on licensing the AI algorithms for personalized Parkinson's disease management, potentially integrating with existing wearable sensors or developing a companion diagnostic to identify suitable patients.

https://pubmed.ncbi.nlm.nih.gov/41853628/

#2

25/50

Paenibacillus dendritiformis C454 as a sustainable biocontrol agent for management of bacterial plant diseases.

Main Researcher: Yael Helman

[AgriTech] [Synthetic Biology] [Other]

"Sustainable bio-pesticide offers powerful, eco-friendly solution to major bacterial plant diseases."

WHY NOW

Rising demand for sustainable agriculture and reduced chemical use drives a rapidly expanding market for effective, environmentally friendly bio-alternatives to traditional pesticides.

COMMERCIAL ANGLE

Commercialization could involve developing a bio-pesticide product based on *P. dendritiformis* C454 for preventative treatment of bacterial plant diseases, addressing the growing demand for sustainable agriculture solutions.

https://europepmc.org/article/MED/41975582

#3

Nitric oxide inhibition ameliorates cortical proteomic changes in the Cntnap2^{-/-} and Shank3^{Î■4-22} mouse models of autism spectrum disorder.

16/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Proteomics] [Clinical]

"Nitric oxide modulation shows promise as a novel therapeutic strategy for autism spectrum disorder."

WHY NOW

Significant unmet medical need exists in ASD, and targeting specific molecular pathways for genetically defined subtypes aligns with precision medicine trends, opening new drug development avenues.

COMMERCIAL ANGLE

Developing a novel therapeutic strategy, potentially a small molecule NO inhibitor or a modulator of NO signaling pathways, could address core biological deficits in a subset of ASD patients with relevant genetic profiles.

https://europepmc.org/article/MED/42010670

#4

Real-world epidemiological outcomes of biologic therapy for chronic rhinosinusitis with nasal polyps: a big data analysis.

24/50

Main Researcher: M Warman

[Clinical] [Immunology] [Software/AI] [Drug Discovery]

"AI-driven diagnostics promise to optimize biologic therapy for chronic rhinosinusitis, reducing healthcare costs."

WHY NOW

Real-world data confirms the efficacy of biologics, creating an immediate need for AI-powered tools to predict patient response, optimize treatment selection, and improve cost-efficiency.

COMMERCIAL ANGLE

Development of companion diagnostics or AI-driven patient stratification tools to predict responsiveness to specific biologics in CRSwNP could optimize treatment selection and further reduce healthcare costs.

https://europepmc.org/article/MED/41486788

#5

Maternal Diabetes and Early Neurodevelopment: Differential Milestone Attainment in Offspring by Diabetes Type.

13/50

Main Researcher: Ofer Beharier

[Neuroscience] [Diagnostics] [Clinical] [Software/AI]

"New insights link maternal diabetes to infant neurodevelopment, opening doors for early diagnostics and interventions."

WHY NOW

The growing prevalence of maternal diabetes necessitates early screening and targeted support to improve long-term neurodevelopmental outcomes for at-risk children.

COMMERCIAL ANGLE

Developing early diagnostic tools or targeted interventions (e.g., specialized therapies, nutritional supplements) for infants at high risk due to maternal diabetes could improve neurodevelopmental outcomes and reduce long-term healthcare costs.

https://europepmc.org/article/MED/41678362

#6

The Natural History of Deficiency of Adenosine Deaminase 2 Vasculitis in a Large Cohort and Factors Associated With Disease-Related Damage.

19/50

Main Researcher: Reeval Segel

[Drug Discovery] [Immunology] [Genomics] [Clinical]

"Biomarkers for DADA2 vasculitis could lead to targeted therapies and improved rare disease management."

WHY NOW

Increased focus on precision medicine for rare diseases, coupled with known efficacy of TNF inhibitors, creates an opportunity for specialized diagnostics and repurposed treatments.

COMMERCIAL ANGLE

The identification of biomarkers associated with DADA2 disease severity and responsiveness to TNF inhibitors presents an opportunity for developing more targeted therapies and companion diagnostics for this rare genetic condition.

https://europepmc.org/article/MED/41539726

#7

Pathways-to-Care Analysis of Non-Hodgkin Lymphoma in Ethiopia: A Mixed-Methods Study.

10/50

Main Researcher: Ora Paltiel

[Clinical] [Software/AI] [Immunology] [Diagnostics]

"Low-cost AI tools could dramatically improve cancer care pathways in resource-limited settings."

WHY NOW

Addressing critical diagnostic delays and treatment abandonment in low-income countries offers a high-impact, scalable solution with global public health significance and social return on investment.

COMMERCIAL ANGLE

Developing low-cost, AI-powered tools for early symptom triage and patient navigation in low-income countries could improve diagnostic rates and reduce delays in cancer care, potentially improving outcomes and reducing mortality.

https://europepmc.org/article/MED/42024844